

Lab 8.1 - Edit VM XML to Add a Network Interface

KVM Virtualisation on Oracle Linux - Day 2, Module 8

Overview

In this lab you will give ol-lab-01 a second network interface entirely by hand-editing its XML definition, rather than using virsh attach-interface. By the end, the VM will have two NICs: its original interface on the bridged network from Lab 4.1, and a new one connected to the default NAT network from Module 2 - giving it both direct LAN reachability and an isolated outbound-only path at the same time.

This lab assumes ol-lab-01 exists and is reachable, and that the default and br0-network networks from earlier labs are both active.

Learning Objective

By the end of this lab, you will be able to:

- Dump and read a VM's current XML definition
- Add a new device block to a VM's XML using virsh edit
- Validate a domain XML file before it's applied
- Confirm a manually added interface is working correctly inside the guest

Step by Step Guide

Prerequisite: Connect to Your Lab VM

Your OCI lab host was provisioned via the Terraform workflow covered earlier in the programme. Before starting this lab, connect to it and confirm ol-lab-01 is present and running.

1. Connect to your lab host over SSH:

```
ssh oracle@<your-lab-instance-ip>
```

Expected result: A shell prompt on your OCI lab instance.

2. Confirm ol-lab-01 exists and is running:

```
virsh list --all
```

Expected result: ol-lab-01 listed with state running. If it's shut off, start it with `virsh start ol-lab-01` before continuing.

Step 1: Dump the Current XML Definition

3. View the VM's current XML, and take note of its existing interface block:

```
virsh dumpxml ol-lab-01 | grep -A5 '<interface'
```

Expected result: One `<interface>` block shown, with `<source network='br0-network'/>` - the bridged network interface from Lab 4.1.

4. Save a copy of the full XML for reference before making any changes:

```
virsh dumpxml ol-lab-01 > ~/ol-lab-01-before.xml
```

Step 2: Add a Second Network Interface

5. Open the VM's live XML for editing:

```
virsh edit ol-lab-01
```

6. Inside the `<devices>` section, alongside the existing `<interface>` block, add a new one connecting to the default network:

```
<interface type='network'>  
  <source network='default'/>  
  <model type='virtio'/>  
</interface>
```

7. Save and exit the editor.

Expected result: Domain ol-lab-01 XML configuration edited. If instead you see an error, the editor reopens with your changes still present - fix the reported issue and save again.

Note: Take care to place the new `<interface>` block inside `<devices>`, as a sibling of the existing one - not nested inside it. Misplaced tags are the most common cause of a validation error at this step.

Step 3: Validate the Change

8. Confirm the saved definition is well-formed:

```
virsh dumpxml ol-lab-01 > ~/ol-lab-01-after.xml
```

```
virt-xml-validate ~/ol-lab-01-after.xml
```

Expected result: No errors printed - the file validates cleanly against libvirt's schema.

9. Confirm two interface blocks now exist:

```
grep -c '<interface' ~/ol-lab-01-after.xml
```

Expected result: 2 printed - one for the original bridged interface, one for the new default-network interface.

Step 4: Reboot and Confirm the New Interface

The new interface was added to the persistent XML, but - like the `--config-only` changes from Module 3 - it won't appear inside the guest until the VM restarts.

10. Reboot the VM:

```
virsh reboot ol-lab-01
```

11. Give it a minute to come back up, then confirm from the host that both interfaces are attached:

```
virsh domiflist ol-lab-01
```

Expected result: Two rows listed: one with Source br0-network, one with Source default, each with its own MAC address.

12. Connect to the console and confirm both interfaces are visible inside the guest:

```
virt-viewer --connect qemu:///system ol-lab-01
```

```
ip addr show
```

Expected result: Two network interfaces listed (beyond loopback), each with its own address - one from the bridged network's segment, one from the 192.168.122.0/24 NAT range.

Outcomes

By the end of this lab, ol-lab-01 has two working network interfaces, added entirely through direct XML editing rather than `virsh attach-interface` - the same underlying change, reached through the layer every other command in this course has been quietly editing on your behalf. You have also practised the safe edit-validate-apply cycle this module covers, which is the same discipline needed for any future change that doesn't have a dedicated `virsh` flag.